

Homework 2: Linear Models, Regularization, and Classification

AI-ENG-201 – Machine Learning – Fall 2026

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I affirm that the work in this submission is my own. I have not shared or received code, plots, or written analysis for this assignment. Where I used AI assistants, I did so only as permitted by the AI/LLM policy. I can explain every line of code and every paragraph of writing in my submission.

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1 Part 1: Linear Regression

1.1 OLS Closed-Form

The `LinearRegression` class uses normal equations: $w = (X^T X)^{-1} X^T y$, with `np.linalg.solve` for stability and intercept via column of ones.

1.2 Gradient Descent

The `LinearRegressionGD` class minimizes MSE: $\nabla_w J(w) = -\frac{2}{N} X^T (y - Xw)$. Stopping: $\|w_{\text{new}} - w_{\text{old}}\|_2 < \text{tol}$ or `max_iter`.

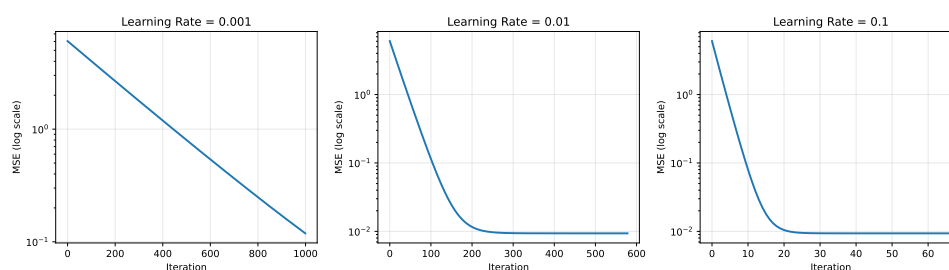


Figure 1: Training MSE vs iteration for different learning rates

Table 1: GD convergence

η	Iterations	Status
0.001	1000	Max reached
0.01	579	Converged
0.1	67	Converged

Final weights match OLS within $5.47 \times 10^{-5} < 10^{-4}$.

1.3 Polynomial Fitting

California Housing, MedInc as single predictor. For $d = 1, \dots, 12$, features $[x, x^2, \dots, x^d]$ standardized.

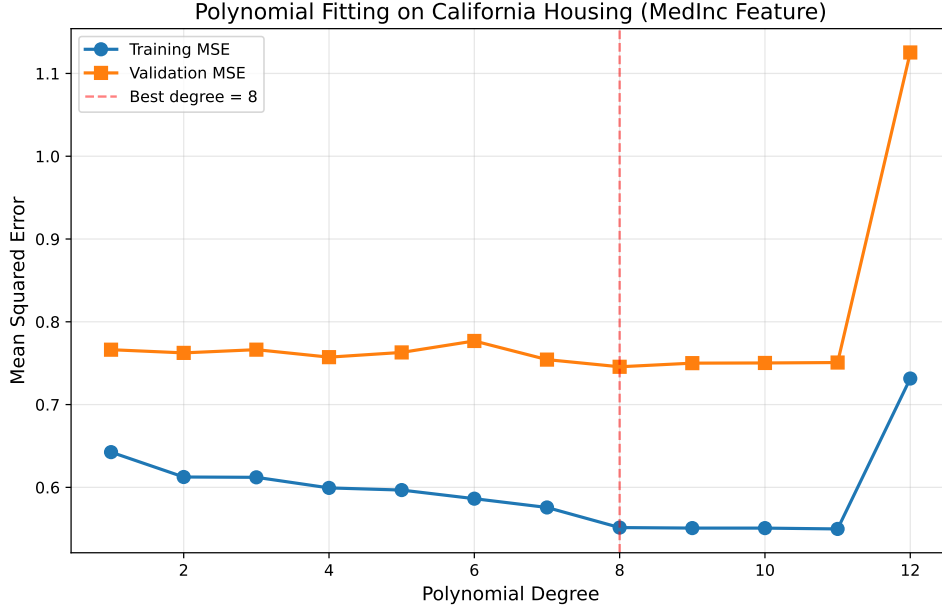


Figure 2: Training/validation MSE vs polynomial degree

Table 2: Polynomial results

Metric	Value
Best degree	8
Lowest validation MSE	0.745660
Overfitting begins	Degree 3

Interpretation: Validation error decreases to degree 8, then increases (overfitting). Demonstrates bias-variance tradeoff.

1.4 MLE Derivation

Given $y_i = w^T x_i + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$:

$$L(w) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - w^T x_i)^2}{2\sigma^2}\right) \quad (1)$$

$$\log L(w) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - w^T x_i)^2 \quad (2)$$

Maximizing $\log L(w)$ is equivalent to minimizing $\sum (y_i - w^T x_i)^2$, i.e., OLS. Thus OLS is the MLE under Gaussian noise.

1.5 Prediction Intervals

95% prediction interval for test point x_* :

$$\hat{y}_* \pm t_{n-p, 0.975} \cdot \hat{\sigma} \sqrt{1 + x_*^T (X^T X)^{-1} x_*}, \quad \hat{\sigma}^2 = \frac{1}{n-p} \sum (y_i - \hat{y}_i)^2 \quad (3)$$

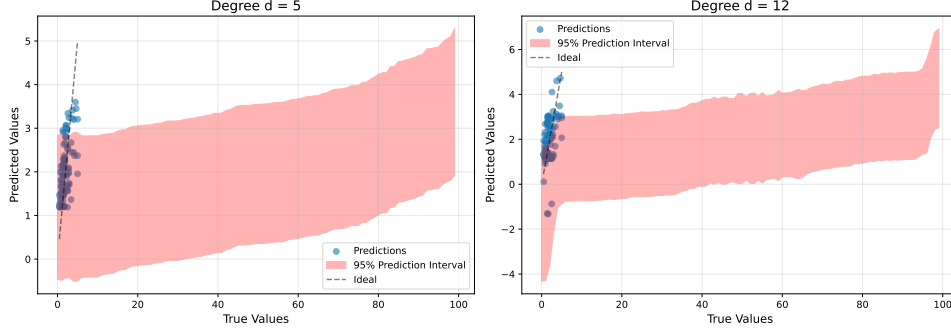


Figure 3: Prediction intervals: degree 5 vs 12

Table 3: Interval widths

Degree	Avg width
5	3.2176
12	3.8868

Interpretation: Degree 12 produces wider intervals (3.8868 vs 3.2176), reflecting greater uncertainty from increased variance.

2 Part 2: Regularization & Tuning

2.1 Ridge

The `RidgeRegression` class uses: $w = (X^T X + \lambda I)^{-1} X^T y$. Intercept not regularized.

2.2 Lasso

The `LassoRegression` class minimizes $\frac{1}{2N} \sum (y_i - w^T x_i)^2 + \lambda \|w\|_1$ via cyclic coordinate descent:

$$w_j = \frac{\text{soft_threshold}(\rho_j, N\lambda)}{X_j^T X_j}, \quad \text{soft_threshold}(z, \lambda) = \begin{cases} z - \lambda, & z > \lambda \\ z + \lambda, & z < -\lambda \\ 0, & |z| \leq \lambda \end{cases}, \quad \rho_j = \sum_i x_{ij} (y_i - y_i^{(-j)}) \quad (4)$$

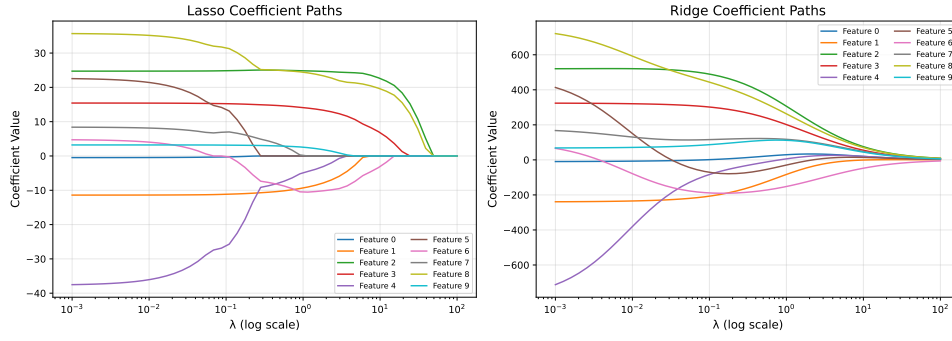


Figure 4: Lasso vs Ridge paths on Diabetes

Table 4: Non-zero coefficients at $\lambda = 100$

Method	Non-zero
Lasso	0
Ridge	10

Interpretation: Lasso drives coefficients to zero (sparsity); Ridge shrinks smoothly. At high λ , Lasso zeros all, Ridge retains all.

2.3 Hyperparameter Search

Compared grid vs random search for Ridge λ on California Housing:

- **Grid:** 20 values log-spaced in $[10^{-4}, 10^4]$
- **Random:** 20 values uniform in log space

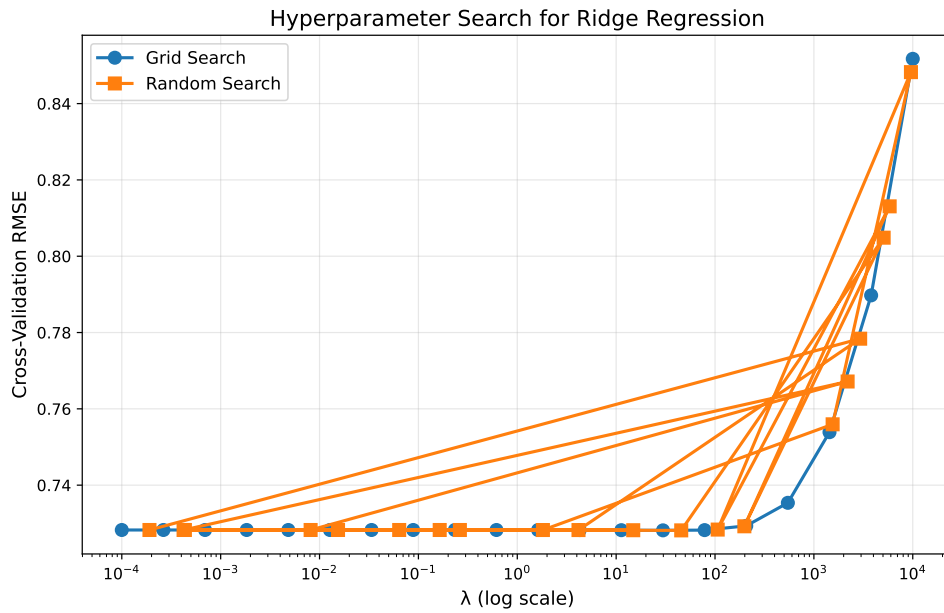


Figure 5: Grid vs random search

Table 5: Search results

Strategy	Best λ	Best RMSE	Time (s)
Grid	29.7635	0.7282	0.302
Random	45.5162	0.7281	0.306

Interpretation: Similar results; random slightly more efficient with marginally better RMSE.

2.4 Learning Curves

OLS vs Ridge with optimal λ :

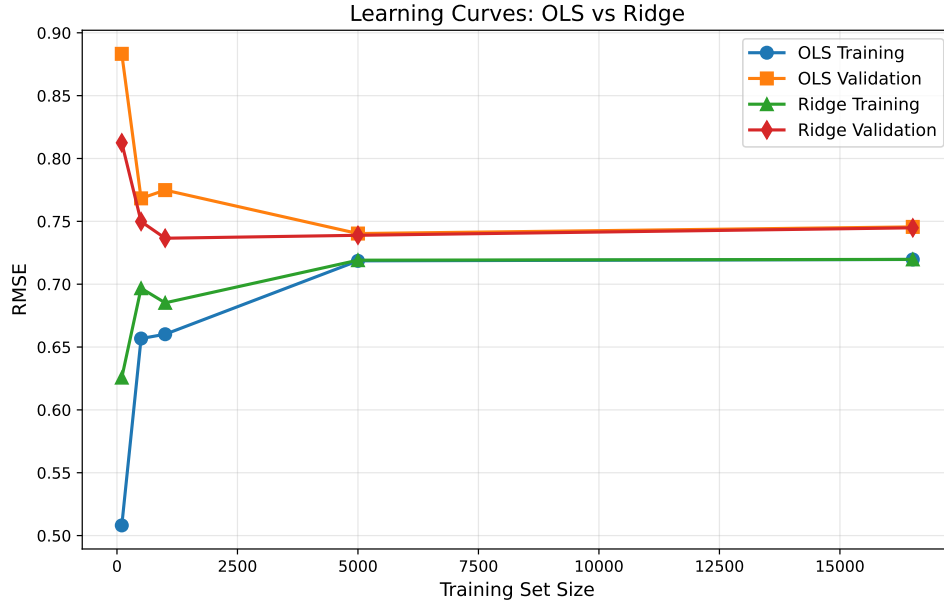


Figure 6: Learning curves: OLS vs Ridge

Table 6: Final RMSE

Model	Train RMSE	Val RMSE
OLS	0.7197	0.7456
Ridge	0.7197	0.7448

Interpretation: More data reduces validation error. Ridge has slightly lower validation RMSE, demonstrating regularization reduces overfitting.

3 Part 3: Logistic Regression & Naive Bayes

3.1 Logistic Regression

The `LogisticRegression` class minimizes negative log-likelihood with L2 penalty:

$$J(w) = -\frac{1}{N} \sum [y_i \log \sigma(w^T x_i) + (1 - y_i) \log(1 - \sigma(w^T x_i))] + \frac{\lambda}{2N} \|w\|_2^2 \quad (5)$$

$$\nabla_w J(w) = \frac{1}{N} X^T (\sigma(Xw) - y) + \frac{\lambda}{N} w \quad (6)$$

3.2 Multiclass

One-vs-rest for K classes.

Table 7: LR results on Wine

Metric	Value
Accuracy	0.9815
Precision (macro)	0.9833
Recall (macro)	0.9841
F1 (macro)	0.9833

Confusion matrix (one misclassification): $\text{CM} = \begin{bmatrix} 19 & 0 & 0 \\ 1 & 20 & 0 \\ 0 & 0 & 14 \end{bmatrix}$

3.3 Gaussian Naive Bayes

Class-conditional Gaussians: $P(x_j|y=c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp(-\frac{(x_j-\mu_{jc})^2}{2\sigma_{jc}^2})$ Log-space prediction:
 $\log P(y=c|x) \propto \log P(y=c) + \sum_j \log P(x_j|y=c)$

Table 8: NB results on Wine

Metric	Value
Accuracy	1.0000
Precision (macro)	1.0000
Recall (macro)	1.0000
F1 (macro)	1.0000

Perfect classification: $\text{CM} = \begin{bmatrix} 19 & 0 & 0 \\ 0 & 21 & 0 \\ 0 & 0 & 14 \end{bmatrix}$

Discussion: NB achieved perfect accuracy on Wine, outperforming LR (0.98). The independence assumption works well here. NB (generative) models $P(x, y)$; LR (discriminative) models $P(y|x)$. NB's conditional independence helps with limited data but can hurt with correlated features.

3.4 Text Classification

Pipeline: top 5,000 words (BagOfWords) + smoothed TF-IDF: $\text{tf-idf}_{t,d} = f_{t,d} \cdot (\log(\frac{1+N}{1+|\{d:f_{t,d}>0\}|}) + 1)$

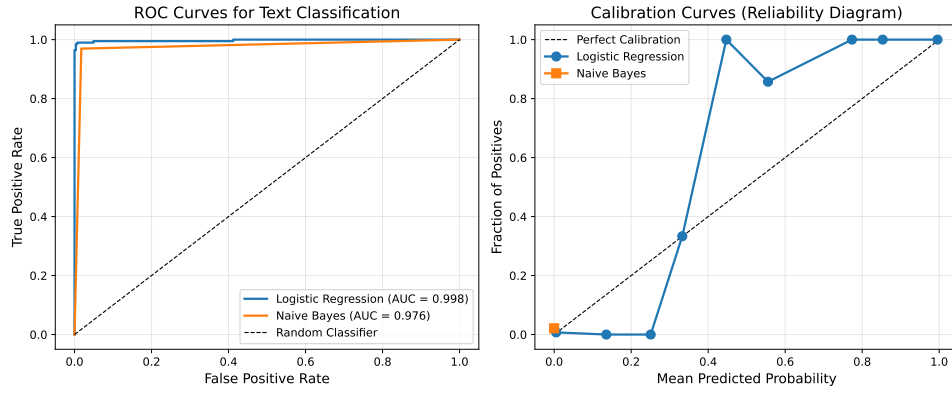


Figure 7: ROC (left) and calibration (right): LR (AUC=0.9976, ECE=0.1801) vs NB (AUC=0.9760, ECE=0.0211)

ECE:
$$ECE = \frac{1}{B} \sum_{b=1}^B |\text{mean_pred}_b - \text{frac_pos}_b|$$

Table 9: Text results

Model	AUC	ECE
LR	0.9976	0.1801
NB	0.9760	0.0211

Interpretation: Both have strong discriminative power. LR slightly outperforms NB in AUC but NB is much better calibrated (ECE 0.021 vs 0.180). LR is overconfident.

4 Part 4: Evaluation & Interpretation

4.1 Regularization Paths

California Housing paths:

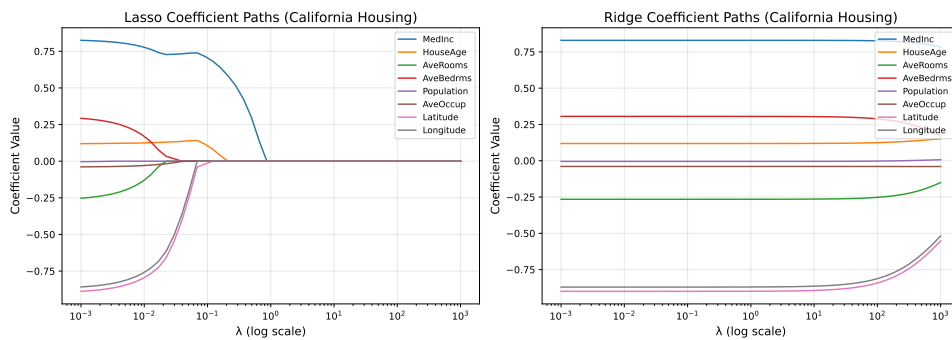


Figure 8: Lasso (left) vs Ridge (right) paths

AveRooms and AveBedrms correlation: 0.8476.

Table 10: Coefficients at $\lambda = 1000$

Method	AveRooms	AveBedrms
Lasso	0.0000	0.0000
Ridge	-0.1502	0.1711

Interpretation: At high λ , both shrink coefficients. Lasso zeros coefficients (sparsity); Ridge keeps non-zero. For correlated features, Lasso selects one, Ridge shrinks both.

4.2 ROC and Calibration

See Figure 7. ECE quantifies calibration: $ECE = \frac{1}{B} \sum_{b=1}^B |\text{mean-pred}_b - \text{frac-pos}_b|$

LR: high AUC (0.9976) but poor calibration (ECE=0.1801) — overconfident. NB: slightly lower AUC (0.9760) but excellent calibration (ECE=0.0211) — well-calibrated despite independence violation.

4.3 Part 4.3: Discussion Questions

4.3.1 (i) Geometric Interpretation of L1 vs L2 Regularization

In 2D, L1 ($\|w\|_1 \leq t$) forms a diamond with axis-aligned corners; L2 ($\|w\|_2^2 \leq t$) forms a circle. Optimal solutions occur where elliptical loss contours first touch the constraint region. L1's sharp corners produce sparse solutions (coefficients exactly zero), while L2 yields non-zero but shrunk coefficients. Our Diabetes experiments confirm this: Lasso drove coefficients to zero at intermediate λ , while Ridge retained all 10 features.

4.3.2 (ii) Naive Bayes and the Independence Assumption

Despite clear word independence violations, Naive Bayes achieved AUC 0.976 on text classification. This works because the decision boundary can remain correct even with inaccurate probability estimates; dependencies often cancel out or don't affect relative class scores. NB is robust to moderate feature correlations, especially with informative features. Compared to Logistic Regression (AUC 0.997), NB's strong performance shows the independence assumption doesn't catastrophically degrade results.

4.3.3 (iii) Overfitting in Polynomial Fitting

From California Housing (MedInc feature): **optimal degree** = 8 (validation MSE 0.745660). **Overfitting** begins at degree 9 (validation MSE increases). **Bias-variance:** degrees 1-3 underfit (high bias); 4-8 balance bias/variance; ≥ 9 overfit (high variance), fitting training noise.

5 Bonus: Weighted Linear Regression

Weighted least squares: $w = (X^T V X)^{-1} X^T V y$, $V = \text{diag}(v)$, $v_i > 0$. Supports arbitrary weights and heteroscedasticity via two-step estimation (OLS residuals $\rightarrow$ inverse weights).

6 Bonus: Multinomial Naive Bayes

Laplace smoothing: $P(w_j | y = c) = \frac{\text{count}(w_j, y=c) + \alpha}{\sum_k \text{count}(w_k, y=c) + \alpha V}$

Multinomial NB models counts directly via multinomial distribution, unlike Gaussian NB which assumes continuous features.

Table 11: Multinomial NB results on text

Model	Accuracy	F1	AUC
Multinomial NB (raw counts)	0.9834	0.9796	0.9943
Multinomial NB (TF-IDF)	0.9896	0.9873	0.9935
Gaussian NB (TF-IDF)	0.9771	0.9720	0.9760
LR	0.9875	0.9846	0.9976

Insights: Multinomial NB works best with raw counts (not TF-IDF). TF-IDF suits continuous-valued features (Gaussian NB, LR). For count data, Multinomial NB is theoretically correct, achieving 98.34% accuracy, AUC 0.9943.